

Model AFR-1 and AFGFR-1

ARC ENERGY REDUCTION with Optional GROUND FAULT, RELAYS

The system is designed for electrical equipment protection, not specifically for personnel protection.

Application: Arc Energy Reduction with (Optional) Ground Fault Protection.

Ground Fault Trip Currents: Ranging from 100-1200A adjustable settings.

For use in Switches with ratings of: 1200A, 1600A, 2000A, 2500A, 3000A, 4000A, 5000A, 6000A.

Arc Energy Reduction: The Arc Energy Reduction section is designed to minimize damage due to Arc Flash events. The relay provides Arc Flash protection only while in Maintenance Mode.

Note: The Arc Energy Reduction section is not designed to prevent the condition, but to minimize the incident energy release and damage due to current spikes.

Ground Fault Detection: These Class 1 Ground Fault relays combined with the proper Boltswitch CT/Sensors, are used for detecting Ground Fault Current in a grounded AC power system.

Note: Use of any other CTs voids the warranty of this product and may result in damage to equipment and the relay.

Input Withstand: 200,000 Amperes RMS for 3 cycles, 50/60 Hz.

Features:

- Meets NEC service entrance equipment standards.
- Power Indication.
- Maintenance Mode Select Switch (w/ connections for remote activation).
- Maintenance Mode Indication.
- Main Trip Contacts: Normally open, Dry Type.
- Auxiliary set of dry contacts, for remote indication.
- Battery powered diagnostic “Trip Result” LED indication feature when relay loses power.
- Operates with bolted pressure contact switches, molded case circuit breakers, or fusible disconnect switches equipped with a shunt trip operator.
- Solid State Contacts provide a faster trip signal for Arc Flash and Ground Fault.
- Positive visual trip indicator.
- “Push-To-Test” primary injection test function for each sensor to verify entire signal path.
- “Shunt-Trip-Bypass” for the main Trip Contacts on AFGF model, with the ability to trip or not trip the shunt trip coil.
- For units with Ground Fault Protection, this feature is active any time the relay is energized regardless of Maintenance Mode status.
- **Arc Flash Section (Uses Rogowski coil sensors):**
 - Arc Flash section uses Rogowski sensors with integral test windings for the “Push-To-Test” function.
 - Sensors connect to relay using plug adapters to minimize wiring issues.
 - Sensors have 6-foot lead wires with connectors.
- **Ground Fault Section (Uses Zero-Sequence Sensor):**
 - Ground Fault section uses Zero Sequence sensor with integral test winding for the “Push-To-Test” function.
 - Trip current ranges available: 100-1200A.
 - Real time fault-current level indicator.
 - Discrete current and time delay threshold settings.
 - Connector for Ground Fault sensor wiring provided with relay.
 - Ground Fault section compatible with any Boltswitch zero-sequence CT(s).

Specifications:

Parameter	Minimum	Maximum
Operating Temperature	30°C	60°C

Parameter	Typical	Maximum	Frequency
Rated power supply	110 Vac	120 Vac	50/60 Hz
AC Power consumption	-	< 2.5 VA	50/60 Hz
AC Voltage Input, Arc Flash Channel	1.0 V (RMS)	1.41 V (RMS)	50/60 Hz
AC Current Input, Ground Fault Channel	0.25 A (RMS)	0.5 A (RMS)	50/60 Hz
AC Main Trip Contacts	120 V (RMS), 5 A (RMS)	240 V (RMS), 10 A (RMS) Make and Carry: 30A for 0.2 sec	50/60 Hz
AC Auxiliary Dry Contacts	120 V (RMS), 1.0 A (RMS)	120 V (RMS), 1.5 A (RMS)	50/60 Hz

Modes of Arc Energy Reduction Operation:

The relay can be placed in 2 modes: Normal (OFF) and Maintenance Mode (ON):

- In Maintenance mode, the relay trips instantaneously when the current exceeds 250% of the switch rating, regardless of the ground fault time delay setting.
- Both the Mode switch on the panel and the remote switch (if used) must be in the “Off” or “Closed” position for the unit to be in “Normal Mode”. If the remote switch is not used, a jumper must be placed across terminals T11 & T12.
- Indications:
 - Maintenance Mode LED (Blue) = Unit in Maintenance Mode
 - Maintenance Mode LED (Off) = Unit in Normal Mode

Arc Energy Reduction: (Only enabled when in Maintenance Mode (ON), and disabled when in Normal Mode (OFF))

The Arc Energy Reduction feature is not designed to prevent, but to minimize damage due to current spikes.

In Maintenance Mode, when the current exceeds the 250% level as indicated in Table 1, the following will occur:

- Main Trip contacts trip instantaneously.
- Auxiliary contacts change state.
- “ARC FLASH” LED turns red if power is still available. If the relay loses power, the “ARC FLASH” red LED indicator can be retrieved by pressing the “Trip Result” button.

Table 1: Trip Level at 250% Overcurrent in Maintenance Mode

Switch Rating (Amps)	Maintenance Mode
	Instantaneous Trip Level (250%)
1200A	3000A
1600A	4000A
2000A	5000A
2500A	6250A
3000A	7500A
4000A	10000A
5000A	12500A
6000A	15000A

For Rogowski Coil Sensor selection see Table 2.

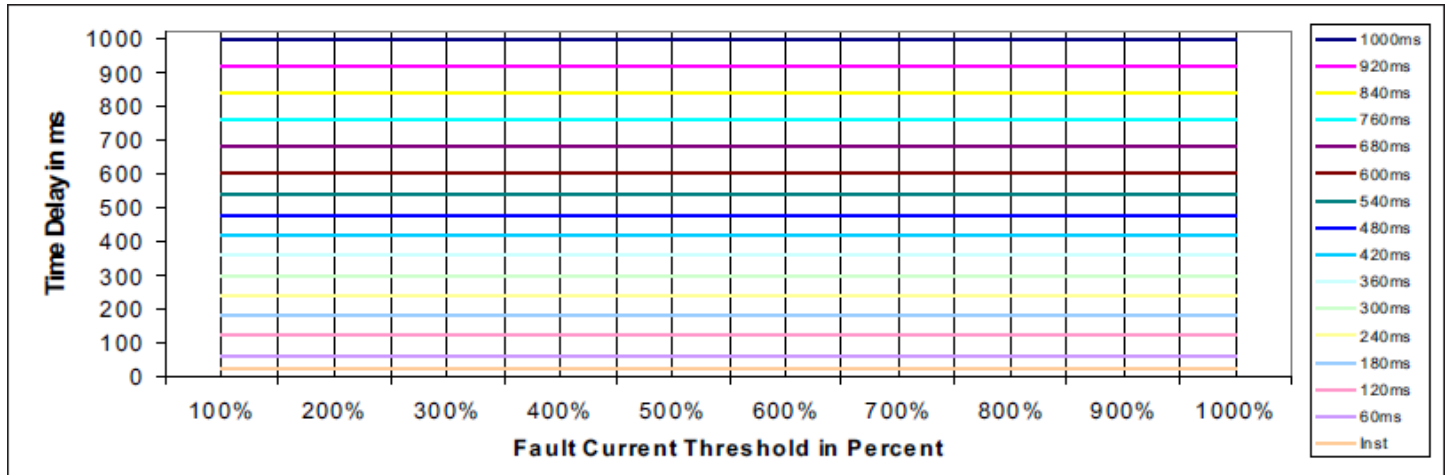
Ground Fault Protection:

The ground fault section has adjustable time delay, adjustable pick-up amps setting, an LED bar graph to indicate ground fault level, and the positive trip indication switch that must be reset manually after the trip.

The relay will indicate the level of the ground fault on the bar graph. In “Normal Mode”, if the ground fault exceeds the set level, “Pick Up Amps”, the ground fault time delay will begin. If the ground fault level remains above the setpoint for the duration of the time delay, the Main Trip contacts trip and Auxiliary contacts will change state. If power is still available, the relay will continue to indicate the level of the ground fault current at the time of the trip. If the relay loses power, the corresponding ground fault trip level indicator can be retrieved by pressing the “Trip Result” button.

Ground Fault Typical Response Curve:

With the Standard Time Delay Curve, the ground fault must be present for the full length of the time delay. The ground fault amperage level does not affect the time delay. (i.e. The time delay will always be as set regardless of the amperage of the ground fault.)



Relay Selection

1. **AFR-1** Arc Flash relay, Solid State Contacts.
2. **AFGFR-1** Arc Flash relay w/ Ground Fault, Solid State Contacts.

ARC FLASH SENSORS

Application:

The Arc Flash Sensors (type AFSRC-) are flexible, Rogowski Coil style used with Boltswitch relays to detect 250% overcurrent conditions. The sensors have integral test windings used for primary injection testing through the entire signal path. Combined with the Boltswitch AFR-1 Arc Flash or, AFGFR-1 Arc Flash/Ground Fault relay, it is the Arc Flash Sensor that determines the Overcurrent Trip Threshold for the relay. Each phase requires a separate Arc Flash sensor.

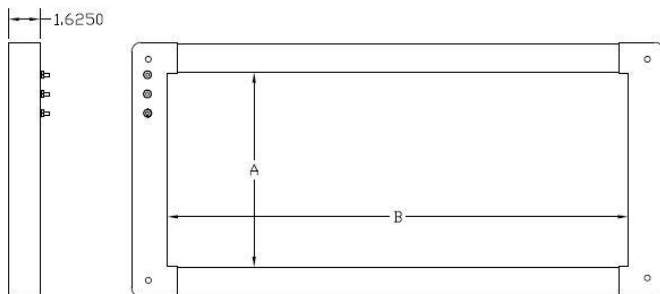
Table 2: Arc Flash Sensor Models

Switch Rating	Sensor Inner Diameter	Sensor Outer Diameter	Part Number
1200A	8"	11"	AFSRC-8
1600A	8"	11"	AFSRC-9
2000A	8"	11"	AFSRC-10
2500A	8"	11"	AFSRC-11
3000A	8"	11"	AFSRC-12
4000A	10"	13"	AFSRC-13
5000A	10"	13"	AFSRC-14
6000A	10"	13"	AFSRC-15

GROUND FAULT SENSORS

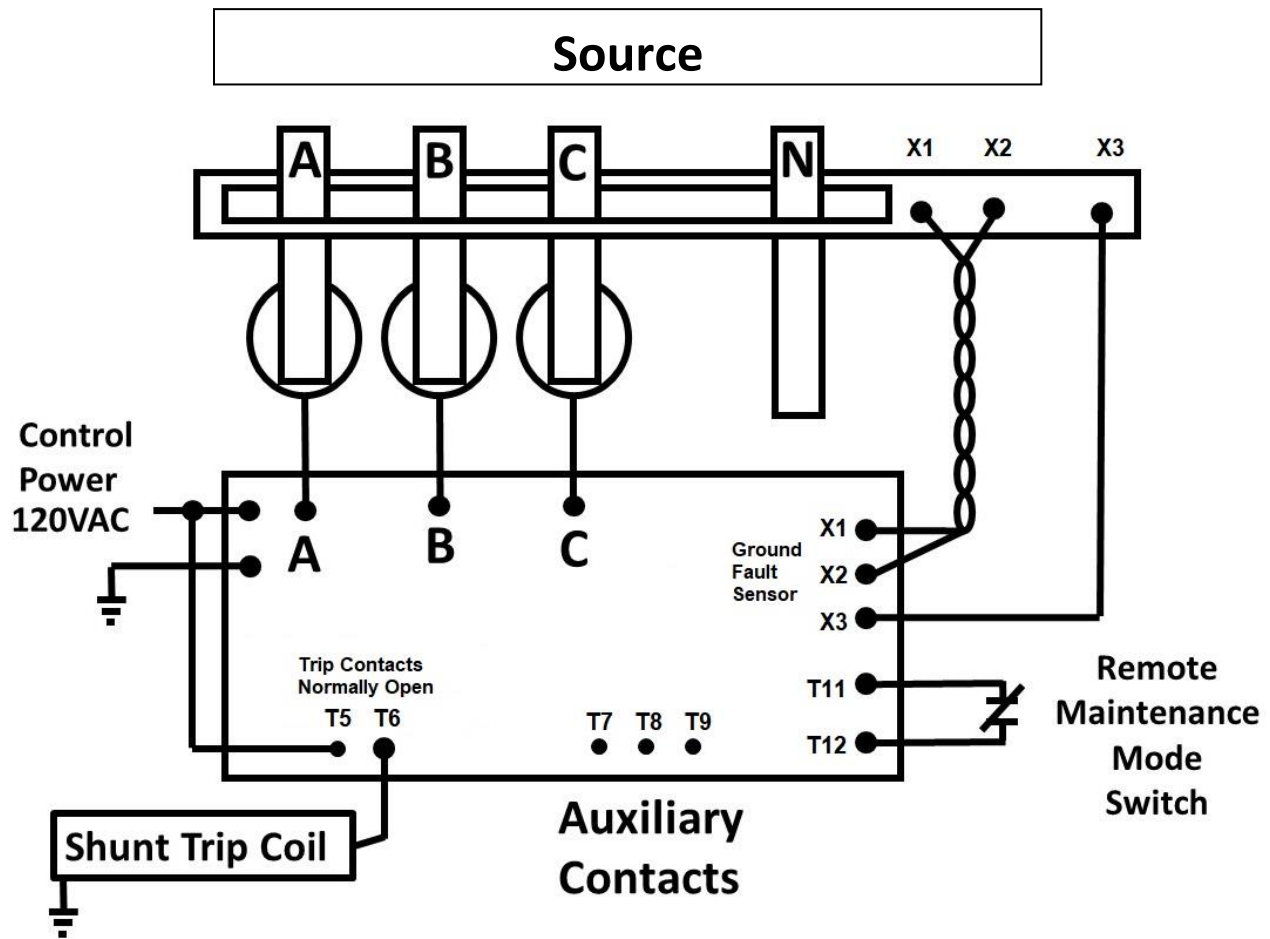
Application:

The Ground Fault Sensors (type GFS-) are available in a variety of sizes. Care should be taken when determining the physical size of the sensor, as all phases plus neutral must pass through the sensor window. The ground fault sensor will only respond to ground faults that occur between the position of the sensor and the downstream load. Each sensor comes with both normal and test windings. Care must be taken to insure the neutral is not bonded anywhere downstream of the ground fault sensor, otherwise nuisance tripping can occur.



CAT. NO.	A	B
GFS-5X36	5"	36"
GFS-10X24	10"	24"
GFS-10X30	10"	30"
GFS-10X36	10"	36"

Typical Installation:



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Notes:

1. The Rogowski sensors can be installed upstream or downstream of the fuses and switch blades.
2. The Trip Contacts and Aux Contacts are dry contacts and isolated from each other.
3. Recommend use of twisted pair for X1 and X2 connections for Ground Fault sensor.
4. Optional remote Maintenance Mode Select Switch (N/C) must be replaced w/ jumper if not used.
5. Arc Flash sensors come with cables and connectors for the relay.
6. When wired integrally from the factory, the grounded conductor is the switched lead.

TESTING

Upon installation of this equipment, it should be tested in accordance with the following instructions. These tests shall be conducted by qualified personnel only.

The proper location of the sensor(s) shall be verified. This can be done visually with knowledge of which bus is involved. The grounding points of the system shall be verified to determine that ground paths which would bypass the ground fault sensor do not exist. The following is a recommended procedure:

1. Disconnect the power source.
2. Disconnect any control transformers and any other devices sensitive to high voltages.
3. Remove the neutral disconnect link.
4. Using a “megger” type meter, measure the resistance to ground of each phase and neutral, if used. No reading of less than 100 ohms is acceptable. Readings of 1 Megohm or more are preferable. Note: High voltage testers and resistance bridges may be used.
5. Reconnect the neutral disconnect link.
6. Reconnect control transformers and other equipment.
7. Reconnect the power source.

Next, test the ground fault protection and/or arc flash protection system to verify compliance with article 230.95(C) and/or 240.67(C) of the National Electrical Code. A simulated or controlled fault current is to be generated and the reaction of the circuit interrupting device observed for correct response. To inject a simulated fault and test for proper grounding of the ground fault components, the following procedure is recommended:

1. Depress and hold the button marked “PTT-GF” on the front of the Relay. The relay should trip instantly or after the set time delay, depending on the system. The interrupting device should trip immediately after the relay trips.
2. Release the button.
3. Reset the Relay (before re-closing the interrupting device).
4. The arc flash portion may also be tested by putting the relay in “Maintenance Mode (ON)” and pushing the PTT-A, PTT-B and PTT-C buttons using the above sequence of steps.

Operation of the ground fault system should be tested periodically and the results recorded. The test record should be held by those in charge of the buildings electrical installation in order to be available to the authority having jurisdiction.

The operation of the ground fault and/or arc flash system can be tested without interrupting service using the following procedure:

1. Depress and hold the button marked “Shunt trip bypass” on the front of the relay.
2. Depress and hold the “PTT-GF” button. The relay should trip instantly or after the set time delay, dependent on the system.
3. Release the “PTT-GF” button.
4. Reset the relay.
5. Release the “Shunt trip bypass” button. **CAUTION:** Failure to first reset the relay will cause the interrupting device to trip upon release of this button.
6. The arc flash feature may also be tested by putting the relay in “Maintenance Mode (ON)” and pushing the PTT-A, PTT-B and PTT-C buttons using the above sequence of steps.

Relay Dimensions

